

1. What is your estimate of the probability $P(\text{EPS exceeded consensus} \mid \text{DriveMed expands})$?

Recall that you updated your probability that last quarter's EPS exceeded the consensus estimate from 45 percent to 82.3 percent after DriveMed announced it would expand. Now you want to update your other priors.

Update your prior probability that DriveMed's EPS met consensus.

Solution:

The probability is $P(\text{EPS met consensus} \mid \text{DriveMed expands}) =$

$$\frac{P(\text{DriveMed expands} \mid \text{EPS met consensus})}{P(\text{DriveMed expands})} P(\text{EPS met consensus})$$

The probability $P(\text{DriveMed expands})$ is found by taking each of the three conditional probabilities in the statement of the problem, such as $P(\text{DriveMed expands} \mid \text{EPS exceeded consensus})$; multiplying each one by the prior probability of the conditioning event, such as $P(\text{EPS exceeded consensus})$; and then adding the three products. The calculation is unchanged from the problem in the text above: $P(\text{DriveMed expands}) = 0.75(0.45) + 0.20(0.30) + 0.05(0.25) = 0.41$, or 41 percent. The other probabilities needed, $P(\text{DriveMed expands} \mid \text{EPS met consensus}) = 0.20$ and $P(\text{EPS met consensus}) = 0.30$, are givens. So

$$\begin{aligned} &P(\text{EPS met consensus} \mid \text{DriveMed expands}) \\ &= [P(\text{DriveMed expands} \mid \text{EPS met consensus}) / P(\text{DriveMed expands})] P(\text{EPS met consensus}) \\ &= (0.20 / 0.41)(0.30) = 0.487805(0.30) = 0.146341 \end{aligned}$$

After taking account of the announcement on expansion, your updated probability that last quarter's EPS for DriveMed just met consensus is 14.6 percent compared with your prior probability of 30 percent.

2. Update your prior probability that DriveMed's EPS fell short of consensus.

Solution:

$P(\text{DriveMed expands})$ was already calculated as 41 percent. Recall that $P(\text{DriveMed expands} \mid \text{EPS fell short of consensus}) = 0.05$ and $P(\text{EPS fell short of consensus}) = 0.25$ are givens.

$$\begin{aligned} &P(\text{EPS fell short of consensus} \mid \text{DriveMed expands}) \\ &= [P(\text{DriveMed expands} \mid \text{EPS fell short of consensus}) / \\ &P(\text{DriveMed expands})] P(\text{EPS fell short of consensus}) \\ &= (0.05 / 0.41)(0.25) = 0.121951(0.25) = 0.030488 \end{aligned}$$

As a result of the announcement, you have revised your probability that DriveMed's EPS fell short of consensus from 25 percent (your prior probability) to 3 percent.